

Computer Architectures

Exam of 3.2.2026 - part I – Group B

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Question #1

The Tomasulo architecture for superscalar processors with dynamic scheduling and speculation uses *Reservation Stations*.

You are requested to

1. Explain what reservation stations are and the role they play in the Tomasulo architecture
2. Describe the fields composing each entry in a reservation station
3. Summarize which modules read and write from/to the reservation stations and when they access them.

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Question #2

Let consider a superscalar RISC-V architecture including the following functional units working in single-precision FP (for each unit the number of clock periods to complete one instruction is reported):

- Integer ALU: 1 clock period
- Data memory: 1 clock period
- FP arithmetic unit: 2 clock periods (pipelined)
- FP multiplier unit: 4 clock periods (pipelined)
- FP divider unit: 8 clock periods (unpipelined)

You should also assume that

- The branch delay slot corresponds to 1 clock cycle, and the branch delay slot is not enabled
- Data forwarding is enabled
- The EXE phase can be completed out-of-order.

You should consider the following code fragment and, filling the following tables, determine the pipeline behavior in each clock period, as well as the total number of clock periods required to execute the fragment. The value of the constants k1 and k2 are written in f10 and f11 before the beginning of the code fragment. Registers x1, x2, x3 and x4 are supposed to store the addresses of vectors v1, v2, v3 and v4.

```
; ***** RISC- *****
; for (i = 0; i < 100; i++) {
;
;     v4[i] = ((v1[i]+k1) / (v2[i]+k2)) * v3[i];
; }
```

```
.data
v1: .word "100 values"
v2: .word "100 values"
v3: .word "100 values"
v4: .word "100 values"
```

```
.text

    addi x5,x0,100
loop: flw f1,0(x1)
      fadd f2,f1,f10
      flw f3,0(x2)
      fadd f4,f3,f11
      fdiv f5,f2,f4
      flw f6,0(x3)
      fmul f7,f5,f6
      fsw f7,0(x4)
      addi x1,x1,4
      addi x2,x2,4
      addi x3,x3,4
      addi x4,x4,4
      addi x5,x5,-1
```

Comments	Clock cycles
x5 ← 100	
f1 ← v1[i]	
f2 ← v1[i]+k1	
f3 ← v2[i]	
f4 ← v2[i]+k2	
f5 ← (v1[i]+k1)/(v2[i]+k2)	
f6 ← v3[i]	
f7 ← (v1[i]+k1)/(v2[i]+k2)*v3[i]	
v4[i] ← f7	
x1 ← x1 + 4	
x2 ← x2 + 4	
x3 ← x3 + 4	
x4 ← x4 + 4	
x5 ← x5 - 1	

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bnez x5,loop
nop

total

branch	

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[illegible]